

COMPUTER SCIENCE**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 11 printed pages.
4. It is divided into two parts: Part I and Part II.
5. It has eleven questions in all.
6. Part I is compulsory and has two questions.
7. While attempting Multiple Choice Questions in Part I, you are required to write only ONE option as the answer.
8. Part II is divided into three sections: A, B and C.
9. Each section in Part II has three questions. Any two questions have to be attempted from each section.
10. The intended marks for questions are given in brackets [].

Instruction to Supervising Examiner

1. Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

PART I – 20 MARKS*Answer all questions.**While answering questions in this Part, briefly indicate your working and reasoning, wherever required.***Question 1**

- (i) According to the Principle of Duality, the Boolean equation $(X' + 0) \cdot (X + Y) = X' \cdot Y$ will be equivalent to: (*Application*) [1]
- (a) $(X' \cdot 1) + (X \cdot Y) = X' + Y$
 - (b) $(X + 1) \cdot (X' + Y') = X \cdot Y'$
 - (c) $(X \cdot 1) + (X' \cdot Y') = X + Y'$
 - (d) $(X' \cdot 0) + (X \cdot Y) = X' + Y$
- (ii) The law which states that $(A')' = A$ is: (*Recall*) [1]
- (a) Complement law
 - (b) Involution law
 - (c) Identity law
 - (d) Idempotent law

- (iii) The complement of the Boolean expression $(A + B') \cdot C$ is: (*Application*) [1]
- (a) $A' \cdot B + C'$
 - (b) $A \cdot B' + C'$
 - (c) $(A' + B) \cdot C'$
 - (d) $A' \cdot B \cdot C'$
- (iv) Study the given propositions and the statements marked Assertion and Reason that follow them. Choose the correct option on the basis of your analysis. [1]
- $p = A$ quadrilateral is a square
 $q =$ All four sides of the quadrilateral are equal
 $s1 = p \rightarrow q$
 $s2 = q \rightarrow p$
- Assertion:** $s2$ is the converse of $s1$.
Reason: $s2$ is true whenever $s1$ is true.
- (*Analysis*)
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
 - (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
 - (c) Assertion is true and Reason is false.
 - (d) Both Assertion and Reason are false.
- (v) The canonical form of the cardinal expression $F(P, Q, R) = \pi(0, 3, 6)$ is: (*Application*) [1]
- (a) $(P + Q + R) \cdot (P + Q' + R') \cdot (P' + Q' + R)$
 - (b) $(P' + Q' + R') \cdot (P' + Q + R) \cdot (P + Q + R')$
 - (c) $P'Q'R' + P'QR + PQR'$
 - (d) $(P + Q + R) \cdot (P' + Q + R') \cdot (P + Q' + R)$
- (vi) Consider the following class declaration, where Vehicle is a class and Electric and Insurable are interfaces: [1]
- `class Car extends Vehicle implements Electric, Insurable`
- Which of the following statements are valid?
- I. Car inherits the non-private members of Vehicle.
 - II. Car must define every abstract method of Electric and Insurable, unless Car is declared abstract.
 - III. Car could also extend a second class along with Vehicle, as `extends Vehicle, Engine`.
- (*Application*)
- (a) I and II
 - (b) II and III
 - (c) I and III
 - (d) Only III

- (vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and choose the correct option. [1]
- Assertion:** The methods `int area(int s)` and `double area(int s)` cannot both be defined in the same class.
- Reason:** Overloaded methods must differ in their parameter lists; the return type alone does not distinguish them.
- (Analysis)
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Both Assertion and Reason are false.
- (viii) A 2-input logic gate gives output 1 only when its two inputs are different. The gate and its expression are: (Understanding) [1]
- (a) XNOR, $A \cdot B + A' \cdot B'$
 (b) XOR, $A \cdot B' + A' \cdot B$
 (c) NAND, $(A \cdot B)'$
 (d) OR, $A + B$
- (ix) State the purpose of the statement `super( . . . )` in the constructor of a sub class. [1]
 (Understanding)
- (x) State the best case and the worst case time complexity of linear search on an array of n elements. (Analysis) [1]

Question 2

- (i) Convert the following infix notation to prefix form. [2]
 $X * (Y - Z) / (P + Q) - R$ (Create)
- (ii) A square matrix $A[m][m]$ is stored in the memory in row major wise, each element requiring 4 bytes of storage. If the address of $A[1][1]$ is 1200 and the address of $A[5][4]$ is 1500, find the order m of the matrix. (Application) [2]
- (iii) The following function `hcf( )` is a part of some class. Assume a and b are positive integers. Answer the questions given below along with dry run / working.

```
int hcf(int a, int b)
{
    if (b == 0)
        return a;
    return hcf(b, a % b);
}
```

- (a) What will the function `hcf( )` return when (i) $a = 84$, $b = 36$ and (ii) $a = 17$, $b = 5$? (Analysis) [2]
- (b) What is the function `hcf( )` performing apart from recursion? (Analysis) [1]

- (iv) The following function `rev( )` is a part of class `Palin`. It returns the reverse of the number `n`; it is first called as `rev(num, 0)`. A number is a palindrome if it is equal to its reverse.
- There are some places in the code marked by ?1?, ?2?, ?3? which may be replaced by a statement / expression so that the function works properly.

```
int rev(int n, int r)
{
    if (n == 0)
        return ?1?;
    return rev(?2?, ?3?);
}
```

- (a) What is the expression or statement at ?1? (*Analysis*) [1]
- (b) What is the expression or statement at ?2? (*Analysis*) [1]
- (c) What is the expression or statement at ?3? (*Analysis*) [1]

PART II – 50 MARKS

Answer six questions in this part, choosing two questions from Section A, two from Section B and two from Section C.

SECTION A

Answer any two questions.

Question 3

- (i) An online grocery store offers free home delivery on an order if any one of the following conditions is satisfied: [5]
- The customer is a prime member and lives within 10 km of the store.
- OR
- The order value is Rs 500 or more, the customer lives within 10 km of the store and it is the festival season.
- OR
- The customer is a prime member living beyond 10 km, whose order value is below Rs 500, during the festival season.

The inputs are:

INPUTS	
P	the customer is a prime member
V	the order value is Rs 500 or more
D	the customer lives within 10 km of the store
F	it is the festival season

(In all the above cases, 1 indicates yes and 0 indicates no.)

Output: X – denotes free delivery [1 indicates YES and 0 indicates NO in all cases]

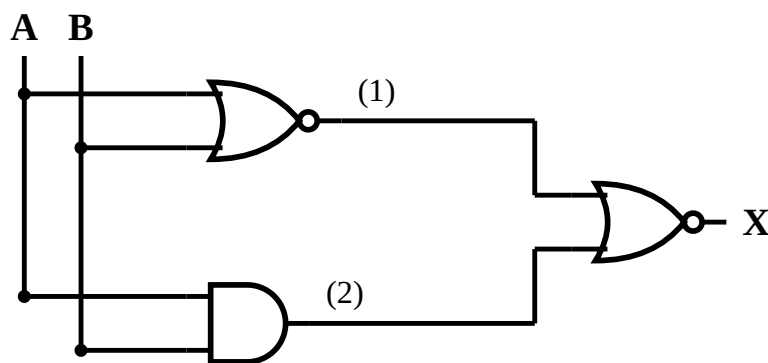
Draw the truth table for the inputs and outputs given above and write the SOP expression for $X(P, V, D, F)$.

(*Application*)

- (ii) Reduce the above expression $X(P, V, D, F)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). Draw the logic gate diagram for the reduced expression using NAND gates only. Assume that the variables and their complements are available as inputs. *(Create)* [5]

Question 4

- (i)
- (a) Reduce the Boolean function $F(A, B, C, D) = (A+B+C+D') \cdot (A+B+C'+D') \cdot (A+B'+C+D') \cdot (A+B'+C'+D') \cdot (A'+B+C+D') \cdot (A'+B+C'+D') \cdot (A'+B'+C+D) \cdot (A'+B'+C'+D)$ by using a 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). *(Application)* [4]
- (b) Draw the logic gate diagram for the reduced expression. Assume that the variables and their complements are available as inputs. *(Create)* [1]
- (ii) From the logic diagram given below:



- (a) Derive the Boolean expressions at (1), (2) and the output X , and reduce X . *(Application)* [3]
- (b) Draw the truth table for X and name the single gate equivalent to the circuit. *(Analysis)* [2]

Question 5

- (i)
- (a) What is an encoder? *(Recall)* [1]
- (b) Draw the logic circuit of a decimal to BCD encoder (inputs D_0 to D_9 , outputs A, B, C, D with A the most significant bit) using OR gates, and write the expression for each output. *(Create)* [4]
- (ii) A majority circuit has three inputs A, B, C and gives output 1 when two or more of its inputs are 1. Draw its truth table, derive the canonical SOP expression, reduce it and draw the logic circuit for the reduced expression. *(Application)* [3]
- (iii) Draw the logic circuit of a 2-input AND gate using NOR gates only. Show the expression at each step. *(Create)* [2]

SECTION B

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem. This can be achieved by using mnemonic names and comments in the program.

(Flowcharts and Algorithms are not required.)

The programs must be written in Java.

Question 6

An **Emirp number** is a prime number which is also prime when its digits are reversed. [10]

Example: 13 and 31 are both prime, so 13 is an Emirp number.

23 is prime but 32 is not, so 23 is not an Emirp number.

Design a class **Emirp** to check if a given number is an Emirp number. Some of the members of the class are given below:

Class name : Emirp

Data members/instance variables:

n : to store the number

rev : to store the reverse of n

Methods / Member functions:

Emirp(int nn) : parameterised constructor to initialise n = nn and rev = 0

void reverse() : to reverse the number n and store it in rev

boolean isPrime(int x, int d) : returns true if x is a prime number, otherwise false, using the **recursive technique**, where d is the divisor being tried (initially 2)

void isEmirp() : checks whether n is an Emirp number by invoking reverse() and isPrime(), and displays an appropriate message

Specify the class Emirp giving details of the constructor(int), void reverse(), boolean isPrime(int, int) and void isEmirp(). Define the main() function to create an object and call the functions accordingly to enable the task.

(Create)

Question 7

A class **ArrOp** holds an array of integers. The array is to be sorted in ascending order using the **selection sort** technique, and then a new object is to be formed containing the elements of the sorted array with all repetitions removed. [10]

Example: INPUT: { 7, 3, 9, 3, 7, 7, 1 }

Sorted array: { 1, 3, 3, 7, 7, 7, 9 }

Without repeats: { 1, 3, 7, 9 }

The details of the members of the class are given below:

Class name : ArrOp

Data members/instance variables:

arr[] : integer array

size : integer to store the number of elements

Member functions/methods:

ArrOp(int s) : parameterised constructor to assign size = s and create the array

void input() : to accept the elements of the array

void selSort() : to sort the array in ascending order using the selection sort technique

ArrOp removeDup() : returns an object whose array holds the elements of the sorted current array with all duplicates removed; its size equals the number of distinct elements

void display() : to display the array

Specify the class ArrOp giving details of the constructor(int), void input(), void selSort(), ArrOp removeDup() and void display(). Define a main() function to create an object and call the functions accordingly to enable the task.

(Create)

Question 8

In a **Caesar cipher** every letter of a text is replaced by the letter k places after it in the alphabet, wrapping round from Z to A (and from z to a). The case of each letter is kept and all other characters are left unchanged. The encoded text is decoded by shifting each letter $26 - k$ places forward. The key k must be from 1 to 25, otherwise the message "INVALID KEY" is displayed. [10]

Example: INPUT: Hello, World! Key: 3
OUTPUT: Encoded: Khoor, Zruog!
Decoded: Hello, World!

Some of the members of the class are given below:

Class name : Caesar

Data members/instance variables:

txt : to store the original text

enc : to store the encoded text

k : to store the key

Methods / Member functions:

Caesar() : default constructor to initialise the data members with legal initial values

void accept() : to accept the text and the key

char shift(char ch, int s) : returns the letter s places after ch, wrapping round and keeping the case; returns ch unchanged if it is not a letter

void encode() : forms enc by shifting every character of txt by k places using shift()

String decode() : returns the text obtained by shifting every character of enc by $26 - k$ places

void display() : displays txt, enc and the decoded text

Specify the class Caesar giving details of the constructor(), void accept(), char shift(char, int), void encode(), String decode() and void display(). Define the main() function to create an object and call the functions accordingly to enable the task.

(Create)

SECTION C

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem stepwise. This can be achieved by using comments in the program and mnemonic names or pseudo codes for algorithms.

The programs must be written in Java and the algorithms must be written in general / standard form, wherever required / specified.

(Flowcharts are not required.)

Question 9

A text editor remembers the last few actions of the user so that they can be undone. A new action is saved on top of the earlier ones, and *Undo* always removes the most recent action first.

Define a class Undo with the following details:

Class name : Undo

Data members/instance variables:

act[] : array to hold the actions (strings)

size : stores the maximum capacity of the array

top : to point to the index of the most recent action

Methods / Member functions:

Undo(int cap) : constructor to initialise size = cap, top = -1 and to create the array

void push(String a) : to save the action a if possible, otherwise display the message "UNDO LIMIT REACHED"

String pop() : to remove and return the most recent action; if there is none, return "NOTHING TO UNDO"

- (i) Specify the class Undo giving details of the functions void push(String) and String pop(). Assume that the other functions have been defined. The main() function and algorithm need NOT be written. *(Create)* [4]
- (ii) Name the entity described above and state its principle. *(Understanding)* [1]

Question 10

A super class **Point2D** has been defined to store a point in a plane. Define a sub class **Point3D** to store a point in space and to find the distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) , given by

[5]

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

The details of the members of both the classes are given below:

Class name : Point2D

Data members/instance variables:

x, y : to store the coordinates (in decimals)

Methods / Member functions:

Point2D(double a, double b) : parameterised constructor to assign x = a and y = b

void display() : to display the coordinates

Class name : Point3D

Data members/instance variables:

z : to store the third coordinate (in decimals)

Methods / Member functions:

Point3D(double a, double b, double c) : parameterised constructor to assign values to the data members of both the classes

double distance(Point3D p) : returns the distance between the current point and the point p

void display() : displays all three coordinates

Assume that the super class Point2D has been defined. Using the concept of inheritance, specify the class Point3D giving the details of the constructor(double, double, double), double distance(Point3D) and void display().

The super class, main function and algorithm need NOT be written.

(Create)

Question 11

- (i) A linked list is formed from the objects of the class Book. The class structure of the Book is given below: [2]

```
class Book
{
    String title;
    Book link;
}
```

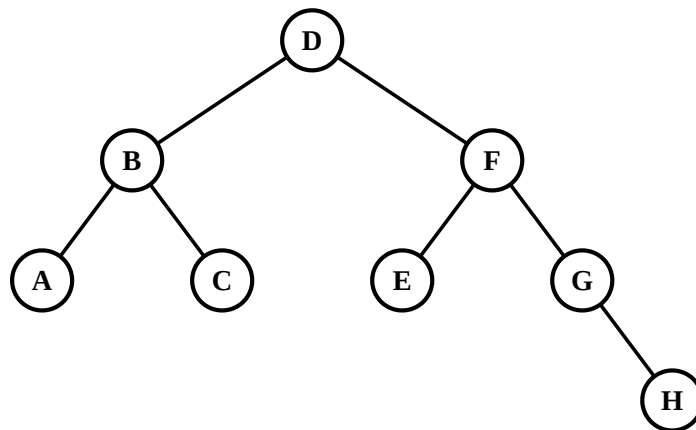
Write an Algorithm OR a Method to add a new node containing the title t at the end of an existing linked list.

The method declaration is as follows:

```
void addBook(Book first, String t)
```

(Analysis)

- (ii) Answer the following questions from the diagram of a Binary Tree given below:



- (a) Write the in-order traversal of the above tree. What does it show about the tree? [1]
(Understanding)
- (b) State, with reason, whether the tree is a full (strictly) binary tree. (Understanding) [1]
- (c) State the degree of node F and name the internal nodes of the right subtree. (Understanding) [1]

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